

ORDINANCE No. 2021-4
Section 23
Solar Power Amendments
To Montgomery County Zoning Ordinance

AN ORDINANCE AMENDING THE MONTGOMERY COUNTY ZONING ORDINANCE
DATED SEPTEMBER 9TH, 1997:

WHEREAS: this article applies to all solar energy installations in the unincorporated areas of Montgomery County, Iowa and

WHEREAS: Montgomery County has adopted this ordinance to meet the goals of the Comprehensive Plan and to preserve the health, safety and welfare of the Community's citizens by promoting the safe, effective and efficient use of active solar energy systems installed to reduce the on-site consumption of fossil fuels, and

WHEREAS: All Montgomery County Ordinances or parts thereof in conflict with the provisions of this Ordinance are repealed, and

WHEREAS: If any section, provision, or part of this Ordinance shall be adjudged invalid or unconstitutional, such adjudication shall not affect the validity of the Ordinance as a whole or any section, provision, or part thereof not adjudged invalid or unconstitutional.

BE IT ENACTED by the Supervisors of the Board of Supervisors of Montgomery County, Iowa, that the following become a part of the Montgomery County Zoning Ordinance dated SEPTEMBER 9TH, 1997

BE IT FURTHER ORDAINED, that this ordinance shall take effect from and after the date that it is passed by the Montgomery County Board of Supervisors, properly signed, published, and has met all other requirements by law.

Effective Date. This Ordinance shall be in effect from and after its final passage, approval, and publication as provided by law.

Date: Introduction & First Reading 7-19-22 Date: Second Reading 7-26-22
Date: Third & Final Reading Waived Date of Publication: 7-13-22

Signed: [Signature], Chairperson Attest: Kathy Burson Deputy, Auditor

Solar Related Definitions

01. **Active Solar Energy System** - A solar energy system whose primary purpose is to harvest energy by transforming solar energy into another form of energy or transferring heat from a collector to another medium using mechanical, electrical, or chemical means.
02. **Building-integrated Solar Energy Systems** - An active solar energy system that is an integral part of a principal or accessory building, rather than a separate mechanical device, replacing or substituting for an architectural or structural component of the building. Building-integrated systems include but are not limited to photovoltaic or hot water solar energy systems that are contained within roofing materials, windows, skylights, and awnings.
03. **Concentrating Solar Power** - (also called **Concentrated Solar Power or CSP**) systems generate power by using mirrors or lenses to concentrate a large area of sunlight, or solar thermal energy, onto a small area. Electricity is generated when the concentrated light is converted to heat, which drives a heat engine (usually a steam turbine) connected to an electrical power generator or powers a thermochemical reaction.
04. **Grid-intertie Solar Energy System** - A photovoltaic solar energy system that is connected to an electric circuit served by an electric utility company.
05. **Ground Cover** – The type of surface treatment to be used on the property to prevent erosion of soil. This may consist of grasses, gravel, asphalt, concrete, or any other options as approved by the Board of Adjustment. A weed control plan shall be included as part of the maintenance plan for the facility.
06. **Ground-mount** – a solar energy system mounted on a rack or pole that rests on or is attached to the ground. Ground-mount systems can be either accessory or principal uses.
07. **Off-grid Solar Energy System** - A photovoltaic system in which the circuits energized by the system are not electrically connected in any way to electric circuits that are served by an electric utility company.
08. **Photovoltaic System** - An active solar energy system that converts solar energy directly into electricity.
09. **Solar Easement, Solar Access Easement** - An easement that limits the height or location, or both, of permissible development on the burdened land in terms of a structure or vegetation, or both, for the purpose of providing access for the benefited land to wind or sunlight passing over the burdened land.
10. **Building/Roof-mount** – a solar energy system mounted on a rack that is fastened to or ballasted on a building roof. Roof-mount systems are accessory to the principal use.

11. **Solar Access** - Unobstructed access to direct sunlight on a lot or building through the entire year, including access across adjacent parcel air rights, for the purpose of capturing direct sunlight to operate a solar energy system.
12. **Utility Scale Solar Installation** - A commercial facility that converts sunlight into electricity, whether by photovoltaics (PV), or other conversion technology, for the primary purpose of wholesale sales of generated electricity. A Utility Scale Solar Installation is the principal land use for the parcel on which it is located.
13. **Community Solar Installation** – A solar-electric (photovoltaic) array that provides electric power to a cluster of households (such as a subdivision) or businesses residing or located offsite from the location of the solar energy system.
14. **Solar Collector** - A device, structure or a part of a device or structure for which the primary purpose is to transform solar radiant energy into thermal, mechanical, chemical, or electrical energy.
15. **Solar Collector Surface** - Any part of a solar collector that absorbs solar energy for use in the collector's energy transformation process. Collector surface does not include frames, supports and mounting hardware.
16. **Solar Energy** - Radiant energy received from the sun that can be collected in the form of heat or light by a solar collector.
17. **Solar Energy System** - A device, array of devices, or structural design feature, the purpose of which is to provide for generation of electricity, the collection, storage and distribution of solar energy for space heating or cooling, daylight for interior lighting, or water heating.
18. **Heat Exchanger** - A component of a solar energy device that is used to transfer heat from one substance to another, either liquid or gas.
19. **Solar Hot Water System** (also referred to as Solar Thermal) - A system that includes a solar collector and a heat exchanger that heats or preheats water for building heating systems or other hot water needs, including residential domestic hot water and hot water for commercial processes.
20. **Mounting Devices** - Frames or other devices that allow the mounting of a solar collector onto a roof surface or the ground.

SOLAR ENERGY STANDARDS

Residential/Small Scale Solar Installations

1. **Permitted Accessory Use.** Active solar energy systems under 50 KW that are allowed as an accessory use are subject to certain requirements as set forth below. Active solar energy systems that do not meet the standards below will require a special use permit.
 - A. **Height.** Active solar energy systems must meet the following height requirements:
 - (1) Building- or roof-mounted solar energy systems shall not exceed the maximum allowed height in any zoning district. For purposes of height measurement, solar energy systems other than building-integrated systems shall be given an equivalent exception to height standards as building mounted mechanical devices or equipment.
 - (2) Ground- or pole-mounted solar energy systems shall not exceed 20 feet in height when oriented at maximum tilt.
 - B. **Set Back.** Active solar energy systems must meet the accessory structure setback for the zoning district and primary land use associated with the lot on which the system is located.
 - (1) **Roof-mounted solar energy systems.** In addition to the building setback, the solar collector surface and mounting devices for roof-mounted solar energy systems shall not extend beyond the exterior perimeter of the building on which the system is mounted or built, unless the solar collector and mounting devices has been explicitly engineered to safely extend beyond the edge, and setback standards are not violated. Exterior piping for solar hot water systems shall be allowed to extend beyond the perimeter of the building on a side yard exposure.
 - (2) **Ground-mounted solar energy systems.** Ground-mounted solar energy systems may not extend into the side-yard or rear setback when oriented at minimum design tilt.
 - C. **Location.**
 - (1) Building integrated or roof-mounted photovoltaic solar energy systems shall be allowed regardless of whether the system is visible from the public right-of-way, provided the building component in which the system is integrated meets all required setback, land use or performance standards for the district in which the building is located.
 - (2) Solar Energy Systems using ground-mounts shall be located in the side-yard or rear-yard in a Residential, Business, Interchange Service, or Industrial district.

- D. Coverage. Roof or building mounted solar energy systems, excluding building-integrated systems, shall allow for adequate access to the roof upon which the panels are mounted to allow for maintenance and inspection of the panels.
 - E. Historic Buildings. Solar energy systems on buildings within designated historic districts or on locally designated historic buildings (exclusive of State or Federal historic designation) will require a special use permit.
 - F. Approved Solar Components. Electric solar energy system components must have a UL listing and solar hot water systems must have an SRCC rating.
 - G. Plan Approval Required. All solar energy systems shall require administrative plan approval by the zoning official.
 - (1) Plan Applications. Plan applications for solar energy systems shall be accompanied by drawings that must show the location of the system on the building, or on the property for a ground-mount system, including the setback from property lines.
 - (2) Plan Approvals. Applications that meet the setback requirements of this ordinance, and do not require a special use permit, shall be granted administrative approval by the zoning official. Plan approval does not indicate compliance with Building Code or Electric Code.
 - H. Compliance with Building Code. All active solar energy systems shall be consistent with the State of Iowa Building Code and solar thermal systems shall comply with HVAC-related requirements of the Electric Code.
 - I. Compliance with State Electric Code. All photovoltaic systems shall comply with the Iowa State Electric Code.
 - J. Compliance with State Plumbing Code. Solar thermal systems shall comply with applicable Iowa State Plumbing Code requirements.
 - K. Utility Notification. All grid-intertie solar energy systems shall comply with the interconnection requirements of the electric utility. Off-grid systems are exempt from this requirement.
2. Special Use. The county encourages the installation of productive solar energy systems and recognizes that a balance must be achieved between character and aesthetic considerations and the reasonable desire of building owners to harvest their renewable energy resources. Where the standards 23.1.A, 23.1.B, and 23.1.C cannot be met without diminishing the minimum reasonable performance of the solar energy system, a special use permit may be sought from the Board of Adjustment.
 3. Solar Access. Montgomery County allows for solar resources to be protected consistent with Iowa Statutes.
 - A. Solar Easements Allowed - Montgomery County allows solar easements to be filed, consistent with Iowa State Code 564A7. Any property owner can purchase an easement across neighboring properties to protect access to sunlight. The

easement can apply to buildings, trees, or other structures that would diminish solar access. In situations where the easements are not voluntarily agreed to, the solar access regulatory board may determine whether or not granting an easement is appropriate, consistent with Iowa Statutes 564A.3.

Community or Utility Scale Solar Installations.

1. Concentrating solar power (CSP) systems shall be prohibited.
2. A site plan shall be submitted and reviewed prior to the approval of a Community or Utility Scale Solar Installation and shall require a Special Use Permit.
3. The application for a Community or Utility Scale Solar Installation shall include the following information on the site plan or in narrative form, supplied by the Community or Utility Scale Solar Installation owner, operator or contractor installing the structure(s):
 - A. Number, location and spacing of solar panels/arrays.
 - B. Type of groundcover proposed for the site.
 - C. Planned location of underground or overhead electric lines.
 - D. Project development timeline which indicates how the applicant will inform adjacent property owners and interested stakeholders in the community.
 - E. Interconnection agreement.
 - F. Decommissioning plan.

Site and Structure Requirements

- A. **Setback.** Setbacks for all structures (including solar arrays) must adhere to the minimum principal setback standards for the zoning district where the project is located and a minimum of 100 feet from any occupied residence unless the occupant grants a waiver. Waivers must be submitted no later than the day before the Board of Adjustments hearing. No setbacks shall be required along property lines between two participating landowners. Greater setbacks may be recommended absent a solar access easement agreement. Setbacks for transformers, inverters, and other noise creating equipment shall be 500 feet from any neighboring occupied property, measured at the property line. In addition, audible noises caused by solar farm operation shall not exceed 60 DB measured at the property line of any neighboring parcel with a dwelling at the time the special use permit is approved.
- B. **Screening.** A landscape buffer may be required to be installed and maintained during the life of the operation. Determination of screening requirements will be made by the Board of Adjustment as part of the review and approval process and will be based on adjacent or nearby surrounding land uses and topography.

- C. Utility Connections. Reasonable efforts shall be made to place all utility connections from the solar installation underground, depending on appropriate soil conditions, shape and topography of the site, distance to the connection, or other conditions or requirements.
- D. Grading plan. A grading plan shall be submitted and shall include all proposed changes to the landscape of the site (e.g., clearing, grading, topographic changes, drainage, tree removal, etc.).
- E. Ground Cover. All ground cover options shall be approved by the Board of Adjustments. Apart from access roads, the county would like to encourage the use of animal friendly (domestic and/or wild) plants such as grass, clover, pollinator mix, prairie mix, or some combination of these. A weed control plan shall be included as part of the maintenance plan for the facility.
- F. Glare minimization. All solar panels must be constructed to minimize glare or reflection onto adjacent properties and adjacent roadways and must not interfere with traffic, including air traffic, or create a safety hazard.
- G. Aviation Protection. For solar farms located within 500 feet of an airport or within approach zones of an airport, the applicant must complete and provide the results of the Solar Glare Hazard Analysis Tool (SGHAT) for the Airport Traffic Control Tower cab and final approach paths, consistent with the Interim Policy, FAA Review of Solar Energy Projects on Federally Obligated Airports, or most recent version adopted by the FAA.
- H. Compliance with local, state and federal regulations. Community and Utility Scale Solar Installations shall comply with all applicable local, state, and federal regulations.
- I. Accessory structures. All accessory structures shall be subject to bulk and height regulations of structures in the underlying zoning district.
- J. Floodplain considerations. Utility scale solar installations are maximum damage potential structures and facilities for purposes of the floodplain district regulations.
- K. Signage. All community and utility scale solar installations shall provide the following at all locked entrances.
 - 1. A visible High Voltage warning sign, manufacturer's, operator's or installer's identification signage, must be displayed.
 - 2. Name(s) and phone number(s) for the electric utility provider.
 - 3. Name(s) and phone number(s) for the site operator.
 - 4. The facilities 911 address, GPS coordinates; and
 - 5. A lock box with keys as needed.

6. Fencing/security. A security fence must be installed along all exterior sides of the utility scale solar installation and be equipped with a minimum of one gate and locking mechanism on the primary access side. Security fences, gates and warning signs must be maintained in good condition until the utility scale solar installation is dismantled and removed from the site. Fencing must meet NEC requirements at the time of application.

Decommissioning and site reclamation plan.

- A. The application must include a decommissioning plan that describes: the anticipated life of the utility scale solar installation; the anticipated way the project will be decommissioned; the anticipated site restoration actions; the estimated decommissioning costs in current dollars; and set aside 125% of the estimated amount in an account ensuring that funds will be available for decommissioning and restoration. This decommissioning plan shall be updated every 5 years and the amount of money set aside in the account shall be adjusted to match 125% of the newly estimated cost.
- B. Following a continuous 1-year period in which no electricity is generated, or if substantial action on the project is discontinued for a period of 1 year, the permit holder will have 1 year to complete decommissioning of the utility scale solar installation. Decommissioning shall be completed in accordance with the approved decommissioning plan. The landowner or tenant must notify the County when the project is discontinued.
- C. The Montgomery County Engineer's Office has implemented a road use agreement for wind and solar projects. An approved road use agreement shall be included in the application for the special use permit for each project.
- D. Montgomery County Zoning Officials shall be granted access to the solar site to inspect any possible violations of this ordinance with 48 hours' written notice. The notice shall list the concerns of the Zoning Officials and a representative from the operating company or utility provider shall accompany the Zoning Official on his inspection. If any violation of this ordinance is found during the inspection it shall be documented. The facility operator will have 60 days from the date of the inspection to bring the site back into compliance or the special use permit will be nullified, and facility operations shall cease until it is brought back to compliance. If no corrective action is taken in 120 days, the facility operator shall begin decommissioning and restoration of the site to prepare it to be returned to its previous use.

SOLAR POWER LOCATION AND CONSTRUCTION PERMIT MONTGOMERY COUNTY PLANNING & ZONING

406 W. 4th Street Red Oak, IA 51566
Phone 712-623-5197

PERMIT # _____

Residential/Small Scale Fee: See table on page 2 Community or Utility Scale Fee \$250.00

Type of Application: Residential _____ Community Solar _____ Utility Scale Solar _____

Type of Solar Array: Building Mounted _____ Roof Mounted _____ Ground Mounted _____

Solar Easement with adjoining property owners needed? Yes _____ No _____ (If yes provide copy)

Applicant Name: _____ Phone Number (____) _____ - _____

Site Address _____ City _____ State _____ Zip _____

Contractors Name _____ Phone Number (____) _____ - _____

Address _____ City _____ State _____ Zip _____

Location-Township _____ Range _____ Section _____ 1/4 _____ Parcel ID# _____

Zoning District Designation for the proposed solar power Array: _____

Is the proposed solar power array located within 500 feet of or in the approach zone of an airport? Yes/No

If yes, have you had a FAA glare study performed? Yes/No

In an Unincorporated Community? _____

Size of solar array L _____ W _____ H _____ Number _____ Acreage size _____ Cost \$ _____

Construction Start Date ____/____/____ Estimated Completion Date ____/____/____

Setback distances to property lines- North _____ South _____ East _____ West _____

Indicate if the following information is attached to the Application. Any/all of this information may be requested by the Board of Adjustment during the application review process.

Site Plan:

Number, Location & Spacing of Panels/Arrays	Yes _____ No _____	Not Applicable
Planned Location of Underground or Overhead Lines	Yes _____ No _____	Not Applicable
Project Development Timeline	Yes _____ No _____	Not Applicable
Interconnection Agreement	Yes _____ No _____	Not Applicable
Decommissioning Plan	Yes _____ No _____	Not Applicable
Grading Plan	Yes _____ No _____	Not Applicable
Proposed ground cover type approved	Yes _____ No _____	Not Applicable

FAA Review/Approval if required
Signage
Fencing

Yes _____ No _____ Not Applicable
Yes _____ No _____ Not Applicable
Yes _____ No _____ Not Applicable

I certify under penalty of perjury and pursuant to the laws of the state of Iowa that this application is true and correct.

Signature of Applicant: _____ Date: _____

Residential or Small-Scale Solar Installation: Application fee is found in the table below, to be paid at application submittal.

Paid _____ Date _____

Community or Utility Scale Solar Installation: Application fee is \$250.00, to be paid at application submittal.

Paid _____ Date _____

Approved _____ Disapproved _____ Reason for Denial (if necessary) _____

Board of Adjustment Chairperson:

Date:

Solar Permit Fees	
VALUE OF BUILDING OR STRUCTURE	FEE
To and including \$5,000	\$50
\$5,001 to \$20,000	\$100
\$20,001 to \$40,000	\$120
\$40,001 to \$60,000	\$140
\$60,001 to \$80,000	\$160
\$80,001 to \$100,000	\$180
\$100,001 to \$120,000	\$200
\$120,001 to \$140,000	\$220
\$140,001 to \$160,000	\$240
\$160,001 to \$180,000	\$260
\$180,001 to \$200,000	\$280
Over \$200,000, add \$10 for each additional \$50,000	